



Practice Exam Questions

Big Fat Myth (University of New South Wales)



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Big Fat Myths Practice Exam

To be completed between Week 5 - Week 7

TOTAL NUMBER OF QUESTIONS: 12

QUESTION TYPES: 9 Multiple Choice Questions and 1 Numerical Entry Question (1 mark each) and 2 Long Response Questions (15 marks each)

TOTAL NUMBER OF MARKS: 40

TIME AVAILABLE: 1 hour

Spend 1 minute per Multiple Choice Question and 20 minutes each on long response questions. The rest of the time should be spent checking and finalising your response.

MULTIPLE CHOICE QUESTIONS

Which statement about triglycerides is true?

- ☐ Triglycerides levels of two 70 kg individuals are the same.
- ☐ Triglycerides are an essential lipid. They are not essential. Can survive without them
- ☐ Triglycerides are made up of two fatty acids and a glycerol. Tri = 3
- ☒ Triglycerides are the storage form of fatty acids.
- ☐ Triglycerides lowers the blood pH.

What is the most likely reason for a person's breath to smell like acetone?

- ☐ There are eating food that are too acidic.
 - ☐ They are dehydrated.
 - ☒ They are on a low carbohydrate diet.
 - ☐ They are eating food that are too alkaline.
 - ☐ They are suffering from untreated diabetes.
- Ketone bodies are produced from the lack of glucose in the diet. This can be the result from fasting or starvation or lack of nutrients in the diet. Need more carbs
- Is true, but not the most correct

Which of the following lifestyle changes is associated with successful weight loss long-term?

- ☐ **Eating smaller portion sizes.** Body will get used to smaller portion sizes and a good long term option
- ☐ Eating more carbohydrates.
- ☐ Limiting fruit intake.
- ☐ Eating more protein and fat.
- ☐ Sticking to a low carbohydrate diet. Can be correct but not feasible in the long term

If you take a 1 g of nickel and heat it up, the mass of the nickel will:

- ☐ increase until the nickel reaches room temperature again.
- ☐ increase.
- ☐ decrease until the nickel reaches room temperature again.
- ☐ decrease.
- ☐ **not change.** Heating up changes the energy, doesn't change the mass. Mass only change when getting rid of the atoms

Which one of the following statements about carbohydrate, protein, and fat is correct?

- ☐ Fatty acids are used as building blocks for new proteins. Fatty acids are put together to form triglycerides
- ☐ **Excess sugar is converted to fats.**
- ☐ Proteins are joined together to form glycogen. Glycogen is made up from sugars and stored
- ☐ Sugars join with fatty acids to make amino acids. Amino acids are made up of ammonia and carbs
- ☐ Amino acids are joined together to form glycogen molecules. They are formed together to form proteins
Glucose are formed together to form glycogen

A study was conducted to test the effectiveness of an asthma drug for children.

All the following are signs of an unreliable study, EXCEPT:

- ☐ Which one is a reliable study
- ☐ The drug company paid for the clinical trial. Incentive for the drug to work, unreliable
- ☐ The study was conducted on 15 male participants. Needed to be tested on both female and male, need to be children
- ☐ The study was conducted over 2 weeks. Too short of a time frame to determine effects
- ☐ The study was done in mice. Needed to be tested on humans, so this is unreliable
- ☐ **The study was published in a peer-reviewed journal.** Reliable —> people looked and evaluated the study

Which one of the following is an example of catabolism? Larger molecules are broken up into smaller molecules.
Opposite is anabolism

- ☐ **A protein processed to form a carbon skeleton and urea.**
- ☐ Glucose molecules forming glycogen. Bigger molecules are formed
- ☐ Glucose and fructose coming together to form sucrose. Bigger molecules are formed
- ☐ Three fatty acids and glycerol forming a triglyceride. Bigger molecules are formed
- ☐ Amino acids joining to form an enzyme. Bigger molecules are formed

Which one of the following is not part of the digestion process?

- ☐ **Protein synthesis in the large intestines.**
- ☐ Absorption by the small intestines. Digestion still happens in the small intestine
- ☐ Production and release of saliva by the mouth. Saliva breaks down the food
- ☐ Release of bile by the gall bladder. Without bile acids u wont be able to break down fat and absorb it
- ☐ Muscular contractions in the oesophagus. Pushes food down

Which food is the most “fattening”?

Note: assume the carrot is 100 grams in each option.

- ☐ One carrot, whole fried in oil.
- ☐ One carrot, raw and chopped.
- ☐ One carrot, whole and raw.
- ☒ One carrot, whole boiled and fried in oil. Foods that are heavily processed are easier for your body to digest and thus more fattening
- ☐ One carrot, whole boiled.

NUMERICAL ENTRY QUESTION

The BMI of an individual is calculated using the formula below.

$$\text{BMI} = \text{kg/m}^2$$

If Alice is 162.5 cm tall and weighs 76 kg, what is her BMI? Give your answer to one decimal place. Only enter the numerical value into the box below.

LONG RESPONSE QUESTIONS

Amy has gone for a hike in the national park and have accidentally gotten lost. She has no idea where she is and there is no mobile signal. Provided she has access to drinking water, using your understanding of stored energy inside the body, explain how she will be able to survive in the park.

In your answer you should talk about the following:

- Why Amy needs energy.
- The different types of stored energy in the body and their locations.
- Which organs are using the stored energy, how they use it, and why?
- How long Amy will be able to survive in the park and why?

Explain the similarities and differences between saturated and unsaturated fats.

In our answer you should refer to:

- Structure of these fats
- Where they come from
- Melting points
- Health implications